Project 1 Validation and AI Use

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Company: Microsoft Corporation (MSFT)
Project period: September-October 2026
Status: Documentation and source/code checks completed. A local terminal capture remains required before final submission.
Named AI disclosure
Tool Tasks performed How the work was checked
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Codex Pulled and organized information from the Microsoft 10-Ks and helped solve the formulas used in the DCF, peer-P/E, pro-forma, and sensitivity analysis. It also helped organize the written reports and Project 1 documents. I checked reported financial facts against Microsoft's FY2026 Form 10-K and primary links. I reviewed model inputs, formulas, dates, units, accounting checks, and output logic. I retained unresolved limitations rather than treating generated text as a source.
Google Antigravity Used as a second study partner for coursework and model-question clarification. I compared its explanations against the course instructions, the Microsoft filing, and the committed model/report evidence. I did not use an AI response as a financial-data source without checking it.
AI tools supported drafting, organization, and explanation. The reported facts, source selection, calculation conventions, and final conditional conclusion remain subject to my review.
Locked changed-input record
The locked record was saved at 2026-09-29T13:52:57-04:00 before scenario results and is recorded in Git commit caa7f1e.
Independent input Human-authored prediction before run Lower/base/higher design
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Consolidated revenue-growth path Lower growth should reduce FY2031E operating profit, FCFE, and value; higher growth should increase all three. Base path shifted down or up by 2 percentage points in every forecast year.
AI/cloud capex as % of revenue Higher capex should leave FY2031E operating profit unchanged in the simplified model but reduce FCFE and value; lower capex should increase FCFE and value. Base path shifted down or up by 5 percentage points in every forecast year.
Actual scenario output and diagnosis
Driver and case FY2031E operating profit FY2031E FCFE Value/share Diagnosis
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Revenue growth - lower $243,208.6m $158,744.0m $228.37 Direction prediction supported. Value moved less than operating profit because higher revenue also requires incremental investment and later cash flows are discounted.
Revenue growth - base $266,352.1m $166,984.2m $238.72 Base reference.
Revenue growth - higher $291,224.2m $175,575.4m $249.49 Direction prediction supported.
Capex intensity - lower $266,352.1m $186,247.0m $270.90 Lower cash reinvestment increased FCFE and value.
Capex intensity - base $266,352.1m $166,984.2m $238.72 Base reference.
Capex intensity - higher $266,352.1m $147,721.4m $206.53 Prediction supported: capex reduced cash available to equity without changing operating profit in this simplified model.
The runner records a restored-base match within $0.000001, zero balance-sheet gaps in each valid scenario, cash above the $25,000m floor, and no revolver draw. These scenario outputs are documented in the committed Lab 11 sensitivity records.

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Remaining execution evidence

The environment used to assemble the documents does not expose a Python interpreter. Before submission, I must run the repository's capture command in my local VS Code terminal, review the resulting docs/executed-output.txt, then commit it. That file will be the real execution evidence; this disclosure does not claim to replace it.

Source and validation index

- Primary filing: sources/filings/MSFT-FY2026-10-K.pdf.
- Locked record: research/lab-11-sensitivity/locked-changed-input-record.md.
- Scenario runner: research/lab-11-sensitivity/msft_sensitivity.py.
- Sensitivity output: research/lab-11-sensitivity/lab-11-submission.md.
- Technical validation: research/lab-11-sensitivity/validation-check.md.